LINUX INTERVIEW PREPARATION GUIDE

CHAPTER

06

Linux Storage and Filesystem Management

Interview-focused notes for DevOps, SysAdmin and Cloud Engineers (1–5 years of experience)

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CHAPTER OBJECTIVE

Understand how Linux manages disks, partitions, filesystems and storage. Learn how to check disk usage, create and mount filesystems, configure persistent mounts, work with LVM and swap, and troubleshoot storage-related problems commonly encountered in production environments.

1 Understanding Linux Storage

Linux servers use storage devices to store operating systems, applications, databases, logs and user data. Common storage devices include:

- **HDD** – Hard Disk Drive.
- **SSD** – Solid State Drive.
- **NVMe** – High-speed storage connected through the NVMe interface.
- **EBS** – Elastic Block Store volumes used with AWS EC2.
- **Network storage** – Storage accessed over a network, such as NFS.

In Linux, storage devices are represented as files under `/dev`.

Device	Description
`/dev/sda`	First SCSI/SATA-style disk
`/dev/sdb`	Second SCSI/SATA-style disk
`/dev/nvme0n1`	NVMe disk
`/dev/xvda`	Common Xen-style virtual disk name
`/dev/sda1`	First partition of `/dev/sda`

Device names depend on the storage controller, kernel and cloud environment. Do not assume every EC2 instance uses the same naming convention.

2 Understanding Disk Partitions

A partition is a logical division of a physical storage device. For example, a 100 GB disk may be divided into:

- Partition 1: 50 GB for the operating system.
- Partition 2: 30 GB for application data.

- Partition 3: 20 GB for backups.

```
# View disks and partitions
lsblk

# Display filesystem information
lsblk -f

# Show partition tables
sudo fdisk -l
```

Example:

```
NAME        SIZE TYPE MOUNTPOINTS
sda          50G disk
├─sda1       49G part /
└─sda2        1G part [SWAP]
sdb         100G disk
```

Here:

- `/dev/sda` is the main disk.
- `/dev/sda1` is mounted at `/`.
- `/dev/sda2` is used as swap.
- `/dev/sdb` is an additional disk.

> **IMPORTANT**
>
> `lsblk` is generally the first command to use when investigating attached disks and partitions.

MBR vs GPT Partition Tables

Feature	MBR	GPT
Maximum disk size	2 TB	Practically unlimited
Number of partitions	4 primary (more using an extended partition)	128 by default
Firmware	Legacy BIOS	UEFI
Reliability	Single partition table	Backup copy of the partition table at the end of the disk

Use GPT for modern systems and for any disk larger than 2 TB. Tools such as `fdisk`, `gdisk` and `parted` can create partitions.

3 Understanding Filesystems

A filesystem determines how data is organized and stored on a disk. Linux supports multiple filesystem types.

Filesystem	Description
ext4	Common Linux filesystem
XFS	High-performance filesystem commonly used on enterprise Linux
Btrfs	Modern filesystem with features such as snapshots
FAT32	Common filesystem for removable media
NTFS	Common Windows filesystem
NFS	Network filesystem used to access remote storage

ext4

Ext4 is widely used on Ubuntu and other Linux distributions.

XFS

XFS is commonly used on RHEL-based systems and is suitable for large filesystems and workloads.

ext4 vs XFS

Feature	ext4	XFS
Default on	Ubuntu, Debian	RHEL, Rocky Linux, Amazon Linux
Grow	Online	Online
Shrink	Offline only	**Not supported**
Grow command	`resize2fs` (uses the device)	`xfs_growfs` (uses the mount point)
Repair tool	`fsck` / `e2fsck`	`xfs_repair`

INTERVIEW TRAP

An XFS filesystem cannot be shrunk. If you need a smaller XFS volume, you must back up the data, recreate the filesystem at the new size and restore the data.

Check Filesystem Type

```
lsblk -f
df -T
```

DEVOPS NOTE

Choosing a filesystem depends on the workload, operating system and operational requirements. Ext4 and XFS are particularly important to understand for Linux server administration.

4 Checking Disk Space and Usage

One of the most common production issues is running out of disk space. Linux provides several commands to investigate storage usage.

df – Check Filesystem Space

```
df -h
```

Displays filesystem usage in human-readable units.

```
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        50G   35G   15G  70% /
```

Column	Meaning
Size	Total filesystem capacity.
Used	Space currently used.
Avail	Available space.
Use%	Percentage of space used.
Mounted on	Mount location.

```
# Check filesystem types
df -Th

# Check inode usage
df -i
```

INTERVIEW NOTE

On ext4, **Used + Avail** is often less than **Size**. By default, ext4 reserves about 5% of the space for the root user, so the system can keep working even when normal users fill the disk.

du – Check Directory Size

`du` helps identify which directories consume disk space.

```
# Check a directory
du -sh /var/log

# Check sizes of directories inside /var
sudo du -h --max-depth=1 /var

# Find the largest directories
sudo du -sh /var/* | sort -h

# Find large files
sudo find /var -type f -size +500M -exec ls -lh {} \;
```

df vs du

Command	Purpose
`df`	Shows filesystem-level space usage
`du`	Shows file and directory-level space usage

PRACTICAL DIFFERENCE

`df` may show a filesystem as full even when `du` cannot account for all the used space. One possible reason is deleted files that are still held open by running processes.

5 Understanding Mounting and Unmounting

Linux uses a single directory hierarchy starting from `/`. A storage device must be mounted to a directory before its filesystem can normally be accessed through that directory. That directory is called a **mount point**.

For example, an additional disk can be mounted at `/data`.

```
# Create a mount point
sudo mkdir -p /data

# Mount a filesystem (assuming /dev/sdb1 already contains a valid filesystem)
sudo mount /dev/sdb1 /data

# Check mounted filesystems
findmnt
df -h

# Unmount a filesystem
sudo umount /data

# Or unmount using the device name
sudo umount /dev/sdb1
```

If unmounting fails with a busy-device error, check which processes are using it:

```
sudo lsof +f -- /data
sudo fuser -vm /data
```

> **WARNING**
> Do not force-unmount a filesystem without understanding which processes are using it and the potential impact.

6 Creating a Filesystem

A newly attached disk or partition may need a filesystem before it can be mounted.

> **WARNING**
> Formatting a device destroys existing filesystem data. Verify the device name carefully before running formatting commands.

Step 1 Identify the disk

```
lsblk
```

Assume the additional disk is `/dev/sdb`.

Step 2 Create a partition

For a simple example, use `fdisk`:

```
sudo fdisk /dev/sdb
```

Command	Purpose
`n`	Create a new partition
`p`	Show partition table
`w`	Write changes and exit
`q`	Exit without saving

For production systems, choose the partition layout carefully before making changes.

Step 3 Create a filesystem

Assuming the new partition is `/dev/sdb1`:

```
# Create an ext4 filesystem
sudo mkfs.ext4 /dev/sdb1

# Or create an XFS filesystem
sudo mkfs.xfs /dev/sdb1
```

Step 4 Mount the partition

```
sudo mkdir -p /data
sudo mount /dev/sdb1 /data

# Verify
df -hT /data
```

7 Persistent Mounting Using /etc/fstab

A filesystem mounted using the `mount` command may not automatically mount after a reboot. To configure persistent mounts, Linux uses `/etc/fstab`, which contains filesystem mounting information.

Example Entry

```
UUID=xxxx-xxxx  /data  ext4  defaults  0  2
```

Field	Meaning
UUID	Unique filesystem identifier
`/data`	Mount point
`ext4`	Filesystem type
`defaults`	Default mount options
`0`	Dump backup setting (normally 0)
`2`	Filesystem check order at boot: `1` = root, `2` = other filesystems, `0` = skip (commonly used for XFS)

Why Use UUID Instead of Device Names?

Device names such as `/dev/sdb1` can change depending on the system's device discovery order. UUIDs provide a more reliable way to identify a filesystem.

```
# Get filesystem UUID
sudo blkid
lsblk -f
```

Configure Persistent Mounting

```
# Open the file
sudo nano /etc/fstab
```

Add the appropriate entry using the filesystem's actual UUID and type. For additional (non-root) disks, add the `nofail` option:

```
UUID=xxxx-xxxx  /data  xfs  defaults,nofail  0  0
```

With `nofail`, the server still boots normally if this disk is missing or detached. Without it, a missing disk can send the server into emergency mode.

Before rebooting, test the configuration:

```
# Check fstab for errors
sudo findmnt --verify

# Reload systemd's view of fstab
sudo systemctl daemon-reload

# Mount everything listed in fstab
sudo mount -a

# Check the mount
findmnt /data
```

IMPORTANT

An incorrect `/etc/fstab` entry can cause boot or mounting problems. Always validate changes before rebooting.

8 Understanding LVM (Logical Volume Manager)

LVM allows Linux storage to be managed more flexibly than traditional fixed partitions. With LVM, storage can be combined into logical volumes and expanded more easily when additional capacity is available.

LVM has three main components:

1. Physical Volume (PV)

A disk or partition initialized for LVM.

```
sudo pvcreate /dev/sdb
```

2. Volume Group (VG)

A storage pool created from one or more physical volumes.

```
sudo vgcreate data_vg /dev/sdb
```

3. Logical Volume (LV)

A virtual block device created from a volume group.

```
sudo lvcreate -L 20G -n app_lv data_vg
```

The relationship is:

```
Physical Volume (PV)
        |
        v
Volume Group (VG)
        |
        v
Logical Volume (LV)
        |
        v
Filesystem
        |
        v
Mount Point (/data)
```

```
# Check LVM information
sudo pvs
sudo vgs
sudo lvs

# Create a filesystem on a logical volume
sudo mkfs.ext4 /dev/data_vg/app_lv

# Mount the logical volume
sudo mkdir -p /data
sudo mount /dev/data_vg/app_lv /data
```

Extend a Logical Volume

Suppose the volume group has available free space. Increase the logical volume by 10 GB:

```
sudo lvextend -L +10G /dev/data_vg/app_lv
```

Then grow the filesystem:

```
# For ext4
sudo resize2fs /dev/data_vg/app_lv

# For XFS, grow the mounted filesystem
sudo xfs_growfs /data
```

The `-r` option combines the logical volume extension and filesystem growth in one step:

```
sudo lvextend -r -L +10G /dev/data_vg/app_lv
```

Extend When the Volume Group Is Full

If the volume group has no free space left, add a new disk to it first:

```
# 1. Initialize the new disk for LVM
sudo pvcreate /dev/sdc

# 2. Add it to the existing volume group
sudo vgextend data_vg /dev/sdc

# 3. Give all free space to the logical volume and grow the filesystem
sudo lvextend -r -l +100%FREE /dev/data_vg/app_lv
```

`-L` specifies an exact size (such as `+10G`), while `-l` specifies extents or a percentage (such as `+100%FREE`).

If you enlarged an existing disk instead (for example, by increasing an EBS volume's size), tell LVM about the new space with `sudo pvresize /dev/sdb`.

> **IMPORTANT**
>
> LVM flexibility does not replace backups. Extending a volume and filesystem requires sufficient available capacity and correct identification of the target device.

9 RAID Basics

RAID (Redundant Array of Independent Disks) combines multiple disks to improve performance, provide redundancy, or both.

Level	Minimum disks	Survives	Main purpose
RAID 0	2	No disk failure	Speed only (striping)
RAID 1	2	1 disk failure	Mirroring
RAID 5	3	1 disk failure	Parity with good usable capacity
RAID 6	4	2 disk failures	Double parity
RAID 10	4	1 disk per mirror pair	Speed and redundancy, common for databases

On Linux, software RAID is managed using `mdadm`:

```
# Check RAID status
cat /proc/mdstat

# Show detailed RAID information
sudo mdadm --detail /dev/md0
```

INTERVIEW LINE

RAID is not a backup.

RAID protects against disk failure, but not against accidental deletion, data corruption or ransomware. Those problems are copied to every disk instantly. You still need separate backups.

CLOUD NOTE

AWS EBS volumes are already replicated within their Availability Zone, so RAID 1 is rarely used on EBS. RAID 0 across multiple EBS volumes is sometimes used to increase performance.

10 Understanding Swap Space

Swap is disk space used as an extension of physical memory. When RAM becomes heavily utilized, Linux may move less-active memory pages to swap.

Swap can be configured as:

- A dedicated swap partition.
- A swap file.

```
# Check current swap usage
free -h
swapon --show
```

Create a Swap File

Example: Create a 2 GB swap file.

```
# Create the file
sudo fallocate -l 2G /swapfile

# Set secure permissions
sudo chmod 600 /swapfile

# Format it as swap
sudo mkswap /swapfile

# Enable swap
sudo swapon /swapfile

# Verify
swapon --show
```

To enable it automatically after reboot, add this entry to `/etc/fstab`:

```
/swapfile none swap sw 0 0
```

What Is Swappiness?

Swappiness controls the kernel's tendency to use swap.

```
# Check its current value
cat /proc/sys/vm/swappiness

# Temporarily change it
sudo sysctl vm.swappiness=10

# Make the change permanent
echo "vm.swappiness=10" | sudo tee /etc/sysctl.d/99-swappiness.conf
sudo sysctl --system
```

A lower value generally makes the kernel less inclined to swap, but the appropriate value depends on workload and system requirements.

> **IMPORTANT**
>
> Swap is not a replacement for sufficient RAM. Heavy swapping can significantly affect application performance.

11 Inodes and Disk Space

Linux filesystems use inodes to store file metadata. An inode stores information such as:

- File permissions.
- Owner and group.
- File size.
- Timestamps.
- References to data blocks.

Each file generally consumes an inode.

```
# Check inode availability
df -i
```

What Happens When Inodes Are Exhausted?

A filesystem may have free disk space but still fail to create new files if all available inodes have been consumed.

This commonly happens when applications generate huge numbers of small files, such as temporary files, cache files or application sessions.

```
# Show inode usage per directory (largest at the bottom)
sudo du --inodes -x /var | sort -n | tail -20

# Alternative: approximate file count grouped by directory path
sudo find /var -xdev -type f | cut -d/ -f1-4 | sort | uniq -c | sort -n
```

For detailed analysis, inspect the directories with the highest counts.

12 Common Storage Troubleshooting Scenarios

Scenario 1 Disk usage reaches 100%

```
# Check filesystem usage
df -h

# Identify large directories
sudo du -sh /var/* | sort -h

# Find large files
sudo find /var -type f -size +1G -exec ls -lh {} \;
```

Investigate logs, backups, application data and temporary files before deleting anything.

Common safe cleanup actions:

```
# Empty a large log file safely (the application keeps writing to it)
sudo truncate -s 0 /var/log/myapp/app.log

# Remove journal logs older than 7 days
sudo journalctl --vacuum-time=7d

# Clear the package cache
sudo apt clean          # Ubuntu
sudo dnf clean all      # RHEL-based
```

For a long-term fix, configure log rotation so logs cannot grow without limit.

Scenario 2 Disk is full but du shows less usage

A deleted file may still be held open by a running process.

```
# Find deleted but open files
sudo lsof +L1
```

If an application is holding a large deleted log file open, arrange a safe restart or log-file reopening procedure. Do not kill processes without understanding their role.

If a restart is not possible, you can free the space immediately using the PID and file descriptor (FD) number shown by `lsof`:

```
sudo truncate -s 0 /proc/<PID>/fd/<FD>
```

Use this only for log-type files; truncating a file the application relies on can break it.

Scenario 3 Disk is attached but not visible at the expected mount point

```
# Check available devices
lsblk

# Check mounted filesystems
findmnt

# Check filesystem signatures and UUIDs
sudo blkid
```

If the device is new, it may need partitioning, formatting and mounting.

Scenario 4 Filesystem is not mounting after reboot

```
# Check the configuration
cat /etc/fstab

# Validate it
sudo findmnt --verify
sudo mount -a

# Check filesystem UUIDs
sudo blkid

# Check boot and mount-related logs
journalctl -b
```

Scenario 5 Disk has free space but cannot create files

```
df -i
```

If inode usage is 100%, identify directories containing excessive small files and investigate why they are accumulating.

Scenario 6 Filesystem suddenly became read-only

Applications report *"Read-only file system"* errors when writing.

```
# Confirm the filesystem is mounted read-only (ro)
findmnt -o TARGET,OPTIONS /data

# Look for disk or filesystem errors
sudo dmesg -T | grep -iE "error|ext4|xfs|i/o"
```

When the kernel detects disk or filesystem errors, it remounts the filesystem read-only to prevent further damage.

Check the health of the underlying disk or EBS volume, take a backup if possible, then unmount the filesystem and repair it:

```
# ext4
sudo fsck -y /dev/sdb1

# XFS
sudo xfs_repair /dev/sdb1
```

For the root filesystem, boot into rescue mode first.

WARNING

Never run `fsck` on a mounted filesystem; it can cause serious corruption. Also avoid simply remounting it read-write with `mount -o remount,rw` before the errors are fixed.

Scenario 7 **Server boots into emergency mode after editing /etc/fstab**

A wrong UUID, mount point or filesystem type in fstab can stop the boot. From the emergency shell:

```
# Make the root filesystem writable
mount -o remount,rw /

# Fix or comment out the incorrect line
vi /etc/fstab

# Reload and reboot
systemctl daemon-reload
reboot
```

On a cloud server without console access, use the provider's serial console, or detach the root volume, attach it to another instance and fix fstab there.

Prevention: always run `findmnt --verify` and `mount -a` before rebooting, and use `nofail` for non-critical disks.

13 Linux Storage in AWS

Storage management is especially important when working with AWS EC2 instances. The most common block storage service for EC2 is Amazon EBS (Elastic Block Store). An EBS volume can be attached to an EC2 instance and used as additional storage.

Typical Workflow

1. Create an EBS volume.
2. Attach it to an EC2 instance.
3. Identify the device using `lsblk`.
4. Create a partition if required.
5. Create a filesystem if it is a new, empty volume.
6. Mount it to a directory.
7. Configure `/etc/fstab` (with `nofail`) if persistent mounting is required.

```
lsblk

# Check the volume is empty: output "data" means no filesystem exists
sudo file -s /dev/nvme1n1

sudo mkfs.ext4 /dev/nvme1n1
sudo mkdir -p /data
sudo mount /dev/nvme1n1 /data
df -hT /data
```

WARNING

This example assumes `/dev/nvme1n1` is a new, empty disk. Never run `mkfs` on a volume containing existing data.

Extending an EBS Volume Without Downtime

A very common task is increasing the size of a server's root disk while it keeps running.

Step 1 Increase the volume size in AWS

Modify the EBS volume size using the AWS Console or CLI.

Step 2 Confirm Linux sees the new size

```
lsblk
```

The disk shows the new size, but the partition is still the old size.

Step 3 Grow the partition

```
# Note the space between the disk and the partition number
sudo growpart /dev/nvme0n1 1
```

Step 4 Grow the filesystem

```
# ext4
sudo resize2fs /dev/nvme0n1p1

# XFS (uses the mount point)
sudo xfs_growfs -d /
```

Step 5 **Verify**

```
df -h
```

If the disk uses LVM, run `pvresize` after `growpart`, then `lvextend -r`.

Important AWS Storage Concepts

AWS service	Purpose
EBS	Persistent block storage for EC2, tied to one Availability Zone
Instance Store	Temporary block storage on the physical host; data is lost when the instance stops or terminates
EFS	Managed shared network file storage (NFS), usable by many instances across Availability Zones
S3	Object storage for files, backups and static content

Remember:

- EBS is block storage.
- EFS is managed network file storage.
- S3 is object storage, not a traditional block device.
- Never store important data only on Instance Store volumes.

On Nitro-based EC2 instances, EBS volumes may appear as NVMe devices even when AWS presents them using different attachment names.

14 Common Interview Questions

Q1. **What is a filesystem?**

A filesystem organizes and manages how data is stored and retrieved on a storage device.

Q2. **What is the difference between a disk and a partition?**

A disk is the storage device. A partition is a logical division of that device.

Q3. What is the difference between `df` and `du`?

`df` reports filesystem-level usage. `du` reports the space consumed by files and directories.

Q4. What is mounting in Linux?

Mounting makes a filesystem accessible through a directory called a mount point.

Q5. What is `/etc/fstab`?

It is a configuration file that defines filesystems and their mount settings, commonly used for persistent mounts across reboots.

Q6. Why should UUIDs be used in `/etc/fstab`?

UUIDs identify filesystems more reliably than device names, which may change.

Q7. What is LVM?

Logical Volume Manager provides flexible storage management using physical volumes, volume groups and logical volumes.

Q8. What is the difference between ext4 and XFS?

Both are Linux filesystems. Ext4 is widely used for general-purpose workloads, while XFS is commonly used for large filesystems and enterprise workloads. Both can grow online, but XFS cannot be shrunk.

Q9. What is swap?

Swap is disk space used to hold memory pages when physical RAM is under pressure.

Q10. What is an inode?

An inode stores file metadata and references to the file's data blocks.

Q11. Why can a server have free disk space but still fail to create files?

The filesystem may have exhausted its available inodes.

Q12. How do you troubleshoot a disk-full issue?

Use `df -h` to identify the full filesystem, `du` to find large directories, `find` to locate large files, and `lsof +L1` to identify deleted files still held open by processes.

Q13. How do you extend an LVM logical volume?

Extend the logical volume using `lvextend`, then grow the filesystem using the appropriate tool, such as `resize2fs` for ext4 or `xfs_growfs` for XFS. If the volume group is full, first add a disk using `pvcreate` and `vgextend`.

Q14. What is the difference between EBS, EFS and S3?

EBS provides block storage, EFS provides managed network file storage, and S3 provides object storage.

Q15. What is the difference between MBR and GPT?

MBR supports disks up to 2 TB and 4 primary partitions, and is used with legacy BIOS. GPT supports much larger disks and 128 partitions by default, keeps a backup partition table, and is used with UEFI.

Q16. Why might a filesystem suddenly become read-only?

The kernel remounts a filesystem read-only when it detects disk or filesystem errors, to prevent further damage. Check `dmesg`, verify the underlying disk, then unmount and repair the filesystem using `fsck` or `xfs_repair`.

Q17. A server goes into emergency mode after you edited `/etc/fstab`. How do you fix it?

From the emergency shell, remount root as read-write using `mount -o remount,rw /`, correct or comment out the wrong fstab line, run `systemctl daemon-reload` and reboot. Prevent it by using `findmnt --verify`, `mount -a` and the `nofail` option.

Q18. How do you increase the root disk of an EC2 instance without downtime?

Increase the EBS volume size in AWS, run `growpart` to grow the partition, then grow the filesystem using `resize2fs` for ext4 or `xfs_growfs` for XFS.

Q19. Is RAID a backup?

No. RAID protects against disk failure, but accidental deletion, corruption or ransomware affect all disks at once. Separate backups are still required.

Q20. What is the difference between EBS and Instance Store?

EBS is persistent network-attached block storage that survives instance stops. Instance Store is temporary storage on the physical host, and its data is lost when the instance stops or terminates.

15 Chapter Takeaways

After completing this chapter, you should understand:

- ✓ Linux storage devices and partitions, including MBR vs GPT.
- ✓ Filesystems such as ext4 and XFS, and their key differences.
- ✓ Disk and directory usage using `df` and `du`.
- ✓ Mounting and unmounting filesystems.
- ✓ Creating filesystems using `mkfs`.
- ✓ Persistent mounting using `/etc/fstab` and the `nofail` option.
- ✓ Filesystem UUIDs.
- ✓ LVM: PV, VG and LV.
- ✓ Extending logical volumes and filesystems.
- ✓ RAID levels and why RAID is not a backup.
- ✓ Swap space and swappiness.
- ✓ Inodes and inode exhaustion.
- ✓ Common disk-full, read-only and emergency-mode troubleshooting.
- ✓ AWS EBS, Instance Store, EFS and S3 storage basics, and growing EBS volumes.

PRACTICAL EXERCISES

1. Use `lsblk` to identify disks and partitions.
2. Check filesystem types using `df -Th` .
3. Check disk usage using `df -h` .
4. Find the largest directories under `/var` .
5. Check inode usage using `df -i` .
6. Create a test filesystem on a disposable virtual disk.
7. Mount it at `/data` and verify using `findmnt` .
8. Configure a test persistent mount using UUID in `/etc/fstab` .
9. Create a small LVM volume in a virtual machine and extend it.
10. Check swap usage using `free -h` and `swapon --show` .
11. Validate your `/etc/fstab` using `sudo findmnt --verify` .
12. Increase the size of a test EBS volume and grow its filesystem using `growpart` .

NEXT CHAPTER

Chapter 07: Linux Networking

You will learn IP addressing, DNS, routing, ports, sockets, firewall basics, network troubleshooting commands and how to investigate connectivity problems between Linux servers and applications.

END OF CHAPTER 6